

Assignment 1

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Assignment 1

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Exercise 1 - Error detection by G and H

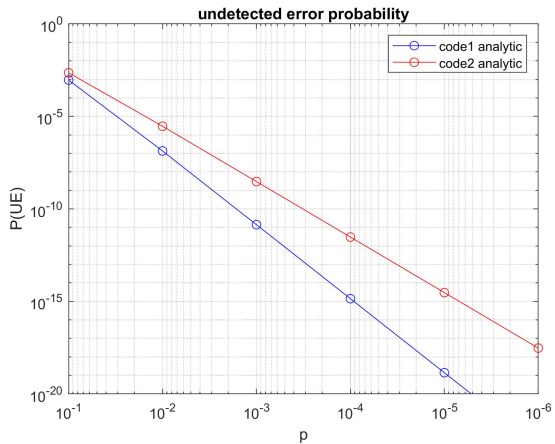
Fix $k = 4$ and $n = 8$. Consider these two generator matrices:

$$G_1 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 & 0 \end{bmatrix}$$

$$G_2 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 & 0 \end{bmatrix}$$

First part = codebook generation

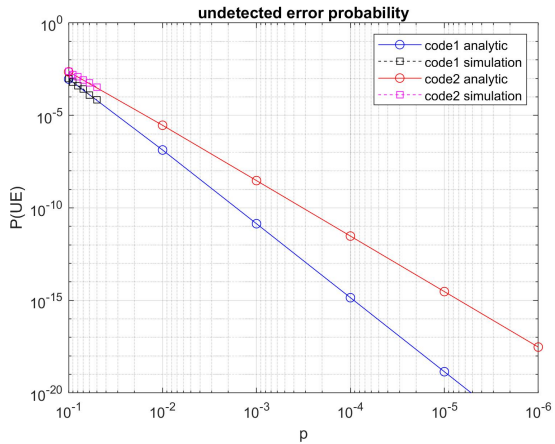
- Generate all the $2^k - 1$ non-zero binary vectors $\underline{v}$ (for example, use `de2bi(i,k)`)
- For each vector generate the codeword $\underline{c} = \underline{v}G_1$
- For each codeword compute the Hamming weight
- Compute A_i (number of codewords with Hamming weight $1 \leq i \leq n$)
- Compute the probability of undetected error $P(UE)$ by using the analytical formula for $p \in \{10^{-1}, 10^{-2}, 10^{-3}, 10^{-4}, 10^{-5}, 10^{-6}\}$
- Plot the curve
- Repeat for the second code



Second part = simulation

Build the parity check matrices H_1 and H_2 . Consider $p_s \in \{0.1, 0.09, 0.08, 0.07, 0.06\}$

- Generate a k -bit random binary information vector (for example, use `randi(2,1,k)-1`)
- Encode it with G_1
- Transmit it over the BSC channel with error probability $p \in p_s$ (for example, use `y=bsc(codeword,p)`)
- Compute the syndrome
- If the syndrome is zero, check if there is an undetected error
- Repeat until you observe at least 100 (if your PC is slow use 50) undetected error events
- Compute the undetected error probability (number of undetected error events/number of transmitted codewords)
- Repeat for all $p \in p_s$ and add the curve to the previous figure
- Repeat for the second code



Third part = discussion

Discuss the results

Exercise 2 - Generator polynomial and code properties:

Consider the primitive polynomial of degree 3: $p(D) = D^3 + D + 1$.

Consider the generator polynomial $g(D) = p(D)(D + 1)$.

Start from $k = 2$ and increase up to $k = 8$.

- 1 Write a program to encode the entire H^k by using $g(D)$ and compute the code distance profile for the different values of k .
- 2 Comment the results. In particular, discuss why the low weight codewords have the observed multiplicity values.

Matlab

```
crc_encoder = comm.CRCGenerator('Polynomial','z^4 + z^3 + z^2 + 1');  
codeword=crc_encoder(v);  
crc_detector = comm.CRCDetector('Polynomial','z^4 + z^3 + z^2 + 1');  
[~,syndrome]=crc_detector(rx_vector);
```

(Last two for the next exercise.)

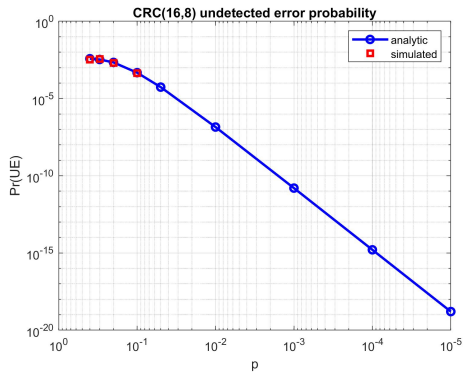
Exercise 3: CRC design and performance evaluation

We want to design a CRC with 8 redundancy bits. The code should be able to neutralize at least the error vectors with weight 1, 2 and 3 for $k \leq 100$.

- 1 Choose the generator polynomial. Discuss why it is able to satisfy the previous condition.
- 2 Fix $k = 8$.
- 3 Encode the entire H^k by using $g(D)$ in a systematic way (for example, use `crc_encoder`).
- 4 Given the codebook, compute the distance profile and discuss the results.

Exercise 3

- 5 Given the distance profile, compute and plot the analytical curve of the undetected error probability vs. p (choose $p \in \{4 \cdot 10^{-1}, 3 \cdot 10^{-1}, 2 \cdot 10^{-1}, 10^{-1}, 5 \cdot 10^{-2}, 10^{-2}, 10^{-3}, 10^{-4}, 10^{-5}\}$).
- 6 Plot the same curve obtained by simulation (choose $p \in \{4 \cdot 10^{-1}, 3 \cdot 10^{-1}, 2 \cdot 10^{-1}, (10^{-1})\}$).



Question 1: distance profile

Given a generic code $C(n, k)$ generated by a linear encoder, prove that:

- 1 The distance profile

$$A_i = |\{\underline{c} \in C : d_H(\underline{c}^*, \underline{c}) = i\}| \quad 0 \leq i \leq n$$

is independent from the codeword $\underline{c}^*$ taken as a reference.

- 2 As a consequence, we can define the distance profile in terms of Hamming weight instead of Hamming distance.

Question 2: Retransmission probabilities

Let us denote by N_{TX} the maximum number of transmissions. For a generic value $N_{TX} = i$:

- 1 Write the expressions of $P(A; C)$, $P(A; UE)$, $P(NA)$.
- 2 Verify that they sum to 1.

Question 3: Parity check matrix H

Given a generic binary linear code $C(n, k)$ with $r = n - k$, generator matrix $G = [I_k \quad P]$ and parity check matrix $H = \begin{bmatrix} P \\ I_r \end{bmatrix}$, we know that the syndrome $\underline{s} = \underline{y}H$ is zero if and only if $\underline{y} \in C$. In this exercise we learn that the parity check matrix of C is not unique.

Suppose that $r = 3$ and denote the matrix by

$$H = [col_1 \quad col_2 \quad col_3]$$

where col_i are column vectors of n bits.

Consider these 3 matrices

$$H_1 = [col_2 \quad col_3 \quad col_1]$$

$$H_2 = [col_1 + col_2 \quad col_2 \quad col_3]$$

$$H_3 = [col_1 + col_2 \quad col_1 + col_2 \quad col_3]$$

- 1 Discuss if these matrices are still admissible parity check matrices of the original code C , i.e., $\underline{s} = \underline{y}H_i$ is zero if and only if $\underline{y} \in C$.
- 2 Discuss if row permutations of H produce an admissible parity check matrix for the original code C .